

Machine Learning Learns a Mapping from Examples

Machine learning (ML) uses examples to estimate a function or discover structure.



CORE CONCEPT

Supervised and Unsupervised Learning Use Different Evidence

Supervised learning uses examples with known targets. The model compares predictions with targets and learns from the error. Unsupervised learning uses data without supplied targets. It searches for structure such as groups, lower-dimensional representations, or unusual observations.

predicting willcancel is supervised;

grouping customers by similar behaviour without a group label is unsupervised.

RUN IT AND INSPECT THE RESULT

Regression Predicts a Quantity

```
actual = 50  
predicted = 46  
residual = actual - predicted  
print(residual)
```

OUTPUT

4

The model under-predicted by 4 minutes.

REASONING SEQUENCE

Classification Predicts a Category

Frame

Binary classification:
two classes, such as
spam/not spam

1

Transform

Multiclass classification:
more than two classes,
such as low/medium/high
risk

2

Validate

A label is a
categorical target
value.

3

Model output: $P(\text{spam} \mid \text{email}) = 0.82$ With threshold 0.50, predict spam because $0.82 \geq 0.50$. With threshold 0.90, predict not spam because $0.82 < 0.90$.

CORE CONCEPT

Clustering Finds Groups Without Labels

Clustering is an unsupervised task that groups observations using a defined similarity or distance. The clusters are discovered from features; no correct cluster label is supplied during training. Clusters are not automatically real customer “types.” Their meaning depends on:

selected features;

scaling;

distance rule;

number or structure of clusters;

stability across samples;

domain interpretation.

[RUN IT AND INSPECT THE RESULT](#)

Training Estimates Parameters; Inference Uses Them

```
model.fit(X_train, y_train)
predictions = model.predict(X_new)
```

OUTPUT

One prediction for each row in X_new.

Training row: features [area=1200, bedrooms=3], target 185000. Inference row: features [area=1350, bedrooms=3], target unknown. The model returns an estimated price using parameters learned earlier.

CORE CONCEPT

Train, Validation, and Test Sets Have Different Jobs

If test results influence repeated model changes, the test set becomes another validation set and no longer provides an independent final estimate. Splits should represent future use. Time-ordered, grouped, or repeated-person data may require specialized splitting instead of random rows.

Training set: estimates model parameters.

Validation set: compares model choices and settings.

Test set: estimates final performance after choices are complete.

training: 700

validation: 150

test: 150

REASONING SEQUENCE

Generalization Is Performance on Unseen Examples

Frame

Underfitting: the model is too limited or poorly trained to capture important structure.

1

Transform

Overfitting: the model learns training-specific noise and performs worse on new data.

2

Validate

Generalization gap: difference between training and validation performance.

3

Model A: training accuracy 70%, validation 69% Model B: training accuracy 99%, validation 74% Model B fits training data much more closely, but its 25-point gap signals overfitting risk.

RUN IT AND INSPECT THE RESULT

A Baseline Sets the Minimum Standard

```
from sklearn.dummy import DummyRegressor

baseline = DummyRegressor(strategy="mean")
baseline.fit(X_train, y_train)
```

OUTPUT

A fitted baseline that predicts the training-target mean.

For test targets [110, 150], predictions are [120, 120]. Absolute errors are [10, 30]; MAE is $(10 + 30) / 2 = 20$.

QUALITY GATE

Leakage Gives the Model Unfair Information



using final payment status to predict whether payment will occur;



calculating scaling values from the complete dataset before splitting;



imputing test values using statistics from test rows;



including duplicate people in both training and test sets.

QUALITY GATE

Evaluation Must Match the Problem



Delivery-time prediction -> regression -> errors in minutes



Fraud detection -> classification -> missed fraud and false alarms have different costs



Customer segmentation -> clustering -> no supplied correct label; evaluate stability and usefulness

GUIDED LAB

Guided Lab: Frame the Problem Before Modelling

01

one observation

02

**features available
at prediction time**

03

**target, if one
exists**

04

**regression,
classification, or
clustering**